PORTFOLIO WEBSITE
enrique-gr.github.io/portfolio

ENRIQUE GARCIA RIVERA

Mechanical Engineering

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STATUS: B.S. Mechanical Engineering, Washington University in St. Louis (Dec 2026)
INTERESTS: Flight Mechanics, GNC, and State Estimation for Unmanned Aerial Systems

ENGINEERING EXPERIENCE

WashU Vertical Takeoff and Landing

Avionics & Systems Integration Lead — Sep 2025 -- Present

- Integrating avionics and power systems for a prototype eVTOL aircraft including flight controller, GPS, IMU, telemetry radios, and distributed power hardware.
- Configuring motors, ESCs, and PX4 parameters while using bench and hardware-in-the-loop checks to diagnose sensor, computing, and communication issues.
- Supporting flight test setup, system monitoring, and flight data logging for post-test analysis and system performance review.

STACK: PX4 • HIL • Sensors • Power Systems • Flight Test

Multiplatform Interactive Robotics Lab, UMKC

Undergraduate Researcher — May 2025 -- Aug 2025

- Built a Teensy 4.0 sensing platform synchronizing IMU and EMG streams at 1 kHz for repeatable laboratory data collection.
- Collected synchronized datasets and developed Python signal processing workflows to characterize sensor noise, bias drift, and timing jitter.
- Supported a MATLAB EKF workflow for drift-reduced motion estimation using IMU and vision-based ground truth data.

STACK: Teensy • MATLAB • Python • IMU • EMG

WashU Design Build Fly

Aerodynamics & Payload Engineer — Sep 2024 -- Jan 2026

- Sized carbon fiber wing spars under 2.5g maneuver loads using ANSYS Static Structural FEA and iterated SolidWorks CAD geometry.
- Designed payload mechanisms and 3D-printed components with GD&T drawings for fabrication and assembly.
- Conducted XFLR5 aerodynamic studies and compared lift and drag predictions across wing configurations to support design selection.

STACK: ANSYS • XFLR5 • SolidWorks • GD&T • Flight Test

TECHNICAL TOOLKIT

COMPUTATION & EMBEDDED SYSTEMS
Python, MATLAB/Simulink, C++, Git, Teensy, Arduino, data acquisition

UAV / MECHATRONICS TOOLS
PX4, HIL testing, IMU/GPS integration, oscilloscopes, multimeters, ESCs, telemetry

MECHANICAL DESIGN & ANALYSIS
SolidWorks, GD&T, ANSYS Mechanical, XFLR5, FEA, fabrication, prototyping

SELECTED PROJECTS

Adaptive Cruise Control

PROJECT

TECHNICAL WORK — MATLAB / Simulink

Modeled vehicle longitudinal dynamics across multiple driving regimes and implemented a PID controller to maintain a two-second following gap. Compared numerical solvers to study transient response and solver stability.

PID CONTROL • ODE45 • EULER • LONGITUDINAL DYNAMICS

Autonomous Navigation Robot

PROJECT

TECHNICAL WORK — C++ / Arduino

Built and programmed a small mobile robot for obstacle avoidance using ultrasonic sensors, H-bridge motor control, and real-time state machine logic. Iterated sensor thresholds and actuator timing through physical hardware testing.

ULTRASONIC SENSING • H-BRIDGE • STATE MACHINE

Sentinel Prototype

PROJECT

TECHNICAL WORK — C++ / Python

Designed and assembled an embedded sensing and control prototype to explore autonomous tracking concepts, multi-sensor feedback, and closed-loop actuation on physical hardware.

MULTI-SENSOR FEEDBACK • CLOSED-LOOP ACTUATION

WebTunnel CFD Demo

PROJECT

TECHNICAL WORK — JavaScript / WebGL

Built a browser-based visualization tool to make fluid-flow concepts more intuitive through interactive feedback, emphasizing rapid iteration and engineering communication.

BROWSER VISUALIZATION • FLUID DYNAMICS • INTERACTIVE

RESEARCH DIRECTION

I am building toward deeper capability in UAV state estimation, flight mechanics, and control. My near-term goal is to understand how sensing, estimator behavior, and control performance interact under noise, drift, delay, and real-world hardware constraints.